

Global Medical Device Supply Chain Analytics

Business Case

1. Background

A global medical device manufacturing company manages procurement, supplier operations, inventory, demand planning, and logistics across multiple facilities. These functions generate large volumes of operational data every day. However, business users rely on multiple spreadsheets and disconnected reports to monitor performance, making it difficult to obtain a unified view of the supply chain.

Without centralized reporting, identifying procurement trends, supplier issues, inventory risks, and logistics costs becomes time-consuming and often delays decision-making.

2. Business Challenge

The organization lacked a centralized Business Intelligence solution capable of providing real-time visibility into key supply chain operations.

As a result:

- Procurement spending was difficult to monitor.
- Supplier performance could not be consistently measured.
- Inventory risks such as stock shortages were not easily identified.
- Forecast accuracy was difficult to evaluate.
- Transportation costs were analyzed manually.
- Executives lacked a single source of truth for operational KPIs.

3. Project Objective

The objective of this project was to design and develop an interactive Power BI dashboard that consolidates supply chain data into a single reporting solution.

The dashboard enables users to monitor procurement performance, evaluate suppliers, analyze inventory, assess demand planning accuracy, and optimize transportation operations through interactive visualizations and KPIs.

4. Business Users

The dashboard is intended for:

- Procurement Managers
- Supply Chain Managers
- Inventory Planners

- Logistics Managers
- Operations Managers
- Executive Leadership

5. Expected Business Benefits

The solution provides several operational benefits, including:

- Centralized reporting across the supply chain.
- Faster access to critical business information.
- Improved supplier performance monitoring.
- Better inventory visibility.
- Data-driven procurement decisions.
- Enhanced logistics cost analysis.
- Executive-level KPI monitoring.

6. Project Scope

The dashboard covers five major business areas:

- Procurement Analytics
- Supplier Performance
- Inventory Health
- Demand Planning
- Transportation Analytics

7. Technology Stack

- Microsoft Power BI
- Power Query
- DAX
- Excel / CSV
- Star Schema Data Model

8. Success Criteria

The project will be considered successful if business users can quickly monitor supply chain performance, identify operational issues, and make informed decisions using a single interactive reporting solution.

Disclaimer: *This project was developed as a personal portfolio project using a simulated supply chain dataset. The company context, business scenarios, and analysis are intended to demonstrate Business Intelligence, data modeling, and dashboard development skills and do not represent actual operational data.*